PYTHON PROJECT DOCUMENTATION

Student Performance Analytics System

Mini Project – EWB Internship

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Course: Python with AI

1. Project Title

Student Performance Analytics System – a mini project completed during the Python with AI CodeAlpha Internship.

2. Objective

The objective of this project is to analyze student performance using Python, NumPy and Pandas. The project calculates total marks, average marks, grades and pass/fail results. It also performs subject-wise analysis and identifies the top-performing students.

3. Technologies Used

- Python
- NumPy
- Pandas
- Visual Studio Code
- GitHub

4. Dataset Description

The dataset contains 20 student records. Each record contains Student ID, Name, Department, marks in six subjects and Attendance. The six subjects are Python, Maths, English, Physics, Chemistry and Biology. The data is stored in the CSV file students.csv.

Field	Description
Student_ID	Unique student ID
Name	Student name
Department	Student department
Python, Maths, English,Physics,Chemistry,Biology	Marks in the six subjects
Physics, Chemistry, Biology	Marks in the three additional subjects
Attendance	Student attendance percentage/value in the dataset

5. Implementation

1. The CSV dataset is loaded using Pandas with pd.read_csv().
2. The data is organized and displayed as a Pandas DataFrame.
3. The program checks the column names, number of students and basic DataFrame information.
4. Total marks are calculated by adding the marks of all six subjects.
5. Average marks are calculated by dividing total marks by six and rounding the result to two decimal places.
6. The calculate_grade() function assigns grades using if-elif-else conditions.
7. A for loop processes each student's average and stores the corresponding grade.
8. The check_result() function determines Pass or Fail using an average of 50 as the passing condition.
9. A second for loop processes each student's result.
10. NumPy is used to calculate the highest average, lowest average, class average and subject-wise averages.
11. Pandas sorting is used to identify the top three students by average marks.

Grading Rule:

Average	Grade
90 and above	A+
80–89.99	A
70–79.99	B
60–69.99	C
50–59.99	D
Below 50	F

Pass/Fail Rule: A student is marked Pass when the average is 50 or above; otherwise the result is Fail.

6. Key Features

- Reads student data from a CSV file.
- Uses a Pandas DataFrame for organizing the data.
- Calculates total and average marks for six subjects.
- Assigns grades using a reusable function.
- Determines Pass/Fail using a reusable function.
- Uses loops for processing student averages.
- Uses NumPy for numerical calculations.
- Calculates highest, lowest and class average.
- Performs subject-wise average analysis.
- Identifies the best subject.
- Identifies the top three students.

7. Output / Screenshots

The following screenshots show the actual output of the completed program.

```
Column Names:
Index(['Student_ID', 'Name', 'Department', 'Python', 'Maths', 'English',
       'Physics', 'Chemistry', 'Biology', 'Attendance'],
      dtype='str')

Number of Students:
20

Basic Information:
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 10 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Student_ID  20 non-null     str
 1   Name        20 non-null     str
 2   Department  20 non-null     str
 3   Python      20 non-null     int64
 4   Maths       20 non-null     int64
 5   English     20 non-null     int64
 6   Physics     20 non-null     int64
 7   Chemistry   20 non-null     int64
 8   Biology     20 non-null     int64
 9   Attendance  20 non-null     int64
dtypes: int64(7), str(3)
```

Figure 1. Column names, number of students and basic DataFrame information.

```
Student Performance:
   Student_ID    Name Department  Python  Maths  English  ...  Biology  Attendance  Total  Average  Grade  Result
0        S001  Dhiraj        BCA      28     82       75  ...       52          75    353    58.83      D    Pass
1        S002    Aman        BCA      65     71       68  ...       62          60    404    67.33      C    Pass
2        S003   Priya        BCA      91     48       94  ...       79          81    492    82.00      A    Pass
3        S004    Neha        BCA      56     62       40  ...       76          70    312    52.00      D    Pass
4        S005   Rohit        BCA      34     39       81  ...       35          52    278    46.33      F    Fail
5        S006  Anjali        BCA      23     36       30  ...       92          87    256    42.67      F    Fail
6        S007   Karan        BCA      48     55       52  ...       64          72    355    59.17      D    Pass
7        S008   Pooja        BCA      89     22       87  ...       81          79    401    66.83      C    Pass
8        S009   Arjun        BCA      67     64       72  ...       61          57    343    57.17      D    Pass
9        S010  Simran        BCA      47     41       39  ...       30          95    230    38.33      F    Fail
10       S011   Vikas        BCA      39     61       57  ...       69          74    371    61.83      C    Pass
11       S012  Muskan        BCA      93     30       95  ...       85          80    482    80.33      A    Pass
12       S013  Aditya        BCA      71     29       74  ...       51          49    354    59.00      D    Pass
13       S014  Shreya        BCA      42     55       80  ...       33          56    300    50.00      D    Pass
14       S015  Manish        BCA      45     51       49  ...       83          45    381    63.50      C    Pass
15       S016   Nisha        BCA      68     73       66  ...       58          63    407    67.83      C    Pass
16       S017  Deepak        BCA      88     83       85  ...       20          78    444    74.00      B    Pass
17       S018   Komal        BCA      61     58       23  ...       19          84    283    47.17      F    Fail
18       S019    Ravi        BCA      75     70       78  ...       21          48    397    66.17      C    Pass
19       S020   Tanya        BCA      36     89       84  ...       71          68    359    59.83      D    Pass
```

Figure 2. Student Performance table with total, average, grade and result.

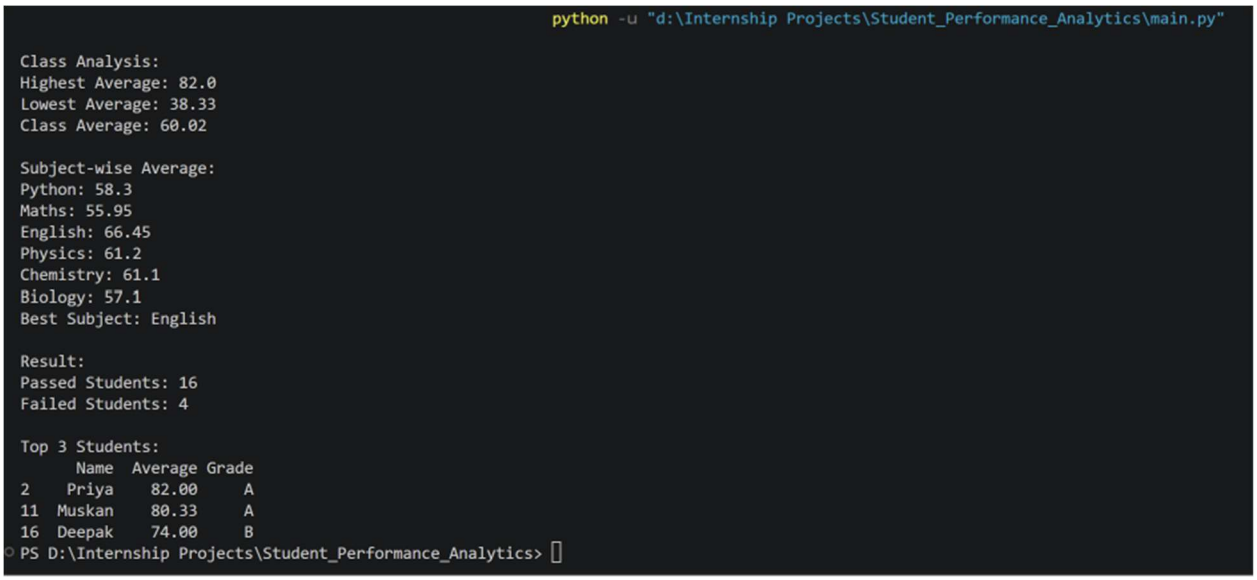


Figure 3. Class analysis, subject-wise averages, best subject, pass/fail count and top three students.

8. Final Outcome

The project successfully analyzed the performance of 20 students. The final output showed:

Analysis	Result
Highest Average	82.00
Lowest Average	38.33
Class Average	60.02
Passed Students	16
Failed Students	4
Best Subject	English

Top 3 Students:

Student	Average	Grade
Priya	82.00	A
Muskan	80.33	A
Deepak	74.00	B

9. Challenges & Learning

During the project, I learned how to read CSV data using Pandas and work with DataFrames. I practiced using functions, loops and conditions to process student records. I also learned how to use NumPy for numerical calculations and how to organize a small Python project.

One practical challenge was updating the calculations after using six subjects instead of three. The total and average calculations were updated accordingly, and the final output was tested to check the results.

10. GitHub Repository

Repository: https://github.com/iamdhiraj-kumar/student-performance-analytics.git

Repository contents: main.py, students.csv and README.md. The project documentation